

Reading Graphs Carefully — Find the Mistake, Then Fix It

Answer Key

Grades 2–3

Quick Answers

1. 8	3. 8	5. 4	7. 5
2. 6	4. 12	6. 5	8. —

Curriculum

Level: Grades 2–3

2.MD / 3.MD – *problems 1, 2, 3, 4, 5, 6*

K.G / 1.G / 2.G / 3.G.A – *problems 7, 8*

Difficulty 1–4 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got 4: counted the steps in the bar and called each step one book, instead of reading the numbers up the side where each step is worth two*
 2. *If they got 3: counted the steps in the bar and called each step one book, instead of reading the numbers up the side where each step is worth two*
 3. *If they got 4: counted the grid steps between the tops of the two bars instead of using the scale, where each step is worth two books*
 4. *If they got 6: counted the steps in the tallest bar instead of reading the number the bar reaches on the side*
 5. *If they got 5: repeated the number printed under the marks instead of counting how many marks were stacked above it*
 6. *If they got 4: answered with how many marks were in the tallest stack instead of the number that stack sits above*
 7. *If they got 6: copied the swapped count, which really belongs to the shape beside it*
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1. Graph A: how many books did Ana read?

Step 1: find what one step is worth before reading any bar. The numbers up the side go 0, 2, 4, 6, 8, 10, 12, so each step is worth 2 books, not 1.

Step 2: Ana's bar reaches the line labelled 8.

Step 3: Ana read 8 books.

Watch for: the answer 4. That is the number of *steps* in the bar. Each step is two books, so four steps is eight books.

2. Find and fix: Nia said Ben read 3 books.

Step 1: name what Nia did. She counted the steps in Ben’s bar and called each step one book. Counting steps is fine — calling each step one book is the mistake.

Step 2: one step is worth 2 books, because the side numbers climb 0, 2, 4, 6 and so on.

Step 3: Ben’s bar is 3 steps tall, and 3 steps of 2 books each is 3×2 , so Ben read 6 books.

Reading the side directly gives the same answer: his bar stops at the line labelled 6.

Full-credit answer says: “one step is 2 books, so Ben read 6.”

3. Graph A: how many more books did Cruz read than Dee?

Step 1: read each bar off the side scale first. Cruz’s bar reaches 12; Dee’s bar reaches 4.

Step 2: “how many more” means subtract: $12 - 4$.

Step 3: Cruz read 8 more books than Dee.

Watch for: the answer 4. That is the number of grid steps between the two bar tops. Four steps of two books each is eight books — the steps have to be turned into books before they mean anything.

4. Find and fix: Sam said the most anyone read is 6.

Step 1: name what Sam did. He counted the steps in the tallest bar. The tallest bar is Cruz’s, and it is 6 steps tall — so Sam’s counting was right and his reading was wrong.

Step 2: each step is worth 2 books, so 6 steps is 6×2 books. Or simply read the side: the bar stops at the line labelled 12.

Step 3: the most books anyone read is 12.

Full-credit answer says: “Sam counted steps, not books.” A quick check: Cruz’s bar is three times as tall as Dee’s, and 12 is three times 4, so the answer is sensible.

5. Find and fix: Rae said 5 boxes had 5 pencils.

Step 1: name what Rae did. She read the number written under the marks and used it twice. On a line plot those two numbers do different jobs: the number under the line says *what* was counted (how many pencils were in the box), and the marks say *how many* boxes were like that.

Step 2: count the marks stacked above the 5: there are four of them.

Step 3: 4 boxes had 5 pencils in them.

Full-credit answer says: “the 5 under the line is the number of pencils; the marks tell how many boxes.”

6. Graph B: which number of pencils happened most often?

Step 1: “happened most often” means find the tallest stack of marks. The stack above 5 has four marks; the stacks above 4, 6 and 7 have two, three and one.

Step 2: the tallest stack is the one above 5. Now answer the question that was actually asked — which *number of pencils*, not how many marks.

Step 3: the number of pencils that happened most often is 5.

Watch for: the answer 4. That is how many marks are in the tallest stack. It answers “how many boxes”, which is a different question.

7. Find and fix: Theo’s side counts.

Step 1: count the straight sides in the picture, shape by shape. A: 3 sides (triangle). B: 4 sides (rectangle). C: 5 sides (pentagon). D: 6 sides (hexagon).

Step 2: compare with Theo’s list. He wrote C as 6 and D as 5 — those two counts are swapped. A and B are correct.

Step 3: shape C has 5 straight sides.

Watch for: the answer 6, which is Theo’s swapped number — it belongs to shape D. Counting corners is a good double-check here: each of these shapes has exactly as many corners as it has straight sides.

8. Show what you know: a four-sided shape that is not a square.

Open task — flagged for manual review. A model answer is below; accept any correct drawing and any equivalent sentence.

(a) Model drawing: a rectangle six grid squares wide and two grid squares tall. It has four straight sides and four square corners, so it is a quadrilateral — but the sides are not all the same length.

Other correct drawings: any rectangle that is longer than it is wide; a parallelogram with slanted corners; a trapezoid; a four-sided shape with four different side lengths.

(b) Model sentence: “A square needs all four sides to be the same length, and my shape has two long sides and two short sides.” A sentence about corners also earns credit if the drawing has a slanted corner: “A square needs four square corners, and mine are slanted.”

What a full-credit answer must contain: a closed drawing with exactly four straight sides that is genuinely not a square, and a sentence that names an *attribute* a square must have — equal sides, or square corners — rather than just saying “it looks different.”